ΣΕΙΡΑ ΑΣΚΗΣΕΩΝ 1: Θεωρία ευστάθειας

ΑΣΚΗΣΗ 1:

Consider the system

$$\dot{x}_1 = x_2 + cx_1(x_1^2 + x_2^2),$$
$$\dot{x}_2 = -x_1 + cx_2(x_1^2 + x_2^2),$$

where c is a constant.

(a) Find the equilibrium points of the system.

(b) Use the Lyapunov candidate

$$V(x) = \frac{x_1^2 + x_2^2}{2}$$

to examine the stability properties of the equilibrium points in (a) for $c = 0$, $c > 0$, and $c < 0$.

ΑΣΚΗΣΗ 2:

Consider the system

$$\dot{x}_1 = -x_1, \qquad \dot{x}_2 = (x_1x_2 - 1)x_2^3 + (x_1x_2 - 1 + x_1^2)x_2$$

(a) Show that $x = 0$ is the unique equilibrium point.

(b) Show, by using linearization, that $x = 0$ is asymptotically stable.

(c) Show that $\Gamma = \{x \in R^2 \mid x_1x_2 \geq 2\}$ is a positively invariant set.

(d) Is $x = 0$ globally asymptotically stable?

ΑΣΚΗΣΗ 3:

Consider the system

(A) $$\dot{x}_1 = x_1 - x_1^3 + x_2, \qquad \dot{x}_2 = 3x_1 - x_2$$

(B) $$\dot{x}_1 = -\tfrac{1}{2}\tan\left(\frac{\pi x_1}{2}\right) + x_2, \qquad \dot{x}_2 = x_1 - \tfrac{1}{2}\tan\left(\frac{\pi x_2}{2}\right)$$

(a) Find all equilibrium point of the system.

(b) Using linearization, study the stability of each equilibrium point.

(c) Using quadratic Lyapunov functions, estimate the region of attraction of each asymptotically stable equilibrium point. Try to make your estimate as large as possible.

(d) Construct the phase portrait of the system and show on it the exact regions of attraction as well as your estimates.

ΑΣΚΗΣΗ 4:

Consider the system

$$\dot{x} = -a[I_n + S(x) + xx^T]x$$

where a is a positive constant, I_n is the $n \times n$ identity matrix, and $S(x)$ is an x-dependent skew symmetric matrix. Show that the origin is globally exponentially stable.

ΑΣΚΗΣΗ 5:

Consider the system $\dot{x} = f(x)+G(x)u$. Suppose there exist a positive definite symmetric matrix P, a positive semidefinite function $W(x)$, and positive constants γ and σ such that

$$2x^T Pf(x) + \gamma x^T Px + W(x) - 2\sigma x^T PG(x)G^T(x)Px \leq 0, \quad \forall\, x \in R^n$$

Show that with $u = -\sigma G^T(x)Px$ the closed-loop system has a globally exponentially stable equilibrium point at the origin.